Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/14/2020
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Comparative Evaluation of Microbial Transcriptomic Responses to Spaceflight Stress: Elucidating Underlying Molecular Mechanisms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	WLN@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-261-3773
Organization Name:	University of Florida		
PI Address 1:	Space Life Sciences Laboratory		
PI Address 2:	505 Odyssey Way, Room 201-B		
PI Web Page:			
City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	12/19/2016	**End Date:**	12/18/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/18/2019 per NSSC information (Ed., 1/11/2019)		
Key Personnel Changes/Previous PI:	October 2019 report: Michael Morrison obtained Ph.D. degree. October 2018 report: None. October 2017 report: None.		
COI Name (Institution):			
Grant/Contract No.:	NNX17AD51G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial cells exposed to the human spaceflight environment are observed to respond differently to similarly-treated ground controls, a phenomenon dubbed the “spaceflight syndrome.” Over the past 4 decades, a number of experiments have been conducted inside various vehicles (e.g., Shuttle, Mir, ISS (International Space Station)) to understand how microbes sense and respond to stresses of human spaceflight including, but not limited to, microgravity and ionizing radiation. In early studies, various phenotypic outputs were measured, such as: growth rate and yield; virulence; biofilm formation and architecture; and resistance to antibiotics or abiotic stresses. With the advent of the genomics and post-genomics revolutions in biology, there have been initial efforts to understand more fundamental molecular aspects of the spaceflight syndrome, by performing global-scale "-omics" analyses of the transcriptome, proteome, metabolome, etc. of microbes exposed to spaceflight. Such studies have yielded valuable insights into the molecular responses of certain individual microbes to the spaceflight environment; furthermore, datasets from these experiments have been deposited into a common, publicly available database, the GeneLab Data System (GDS). We propose to use the data in the GDS as a starting point to explore commonalities and differences in the responses to spaceflight of various different microbes, with an eye towards uncovering possible common fundamental mechanisms. Working Hypothesis: Exposure to the human spaceflight environment alters the physicochemical environment immediately surrounding bacterial cells and within the cells themselves (mainly due to microgravity effects and ionizing radiation). Cells sense the altered conditions and respond using the systems they possess--which evolved on Earth, not in space. A large proportion of microbial physiological responses are regulated at the level of transcription, thus the transcriptome is a logical place to search for common fundamental mechanisms. Objectives: In order to address this hypothesis, we propose the following Specific Aims: 1. Mine the GDS bacterial transcriptome database and reprocess the data in a fashion that will allow cross-organism comparisons. 2. Search the spaceflight literature to identify additional relevant datasets that may have escaped inclusion in the GDS. Deposit these additional datasets in the GDS. Access and reformat the additional data as in Specific Aim 1. 3. Perform bioinformatics analyses to identify pathways and regulons significantly activated or repressed by exposure to spaceflight vs. ground controls in individual organisms. 4. Compare transcriptomes of the same organism grown in simulated microgravity (clinostats) vs. actual spaceflight conditions. 5. Perform comparisons across species, genera, families, and higher taxonomic groupings to find similarities and differences in spaceflight responses. 6. Develop specific hypotheses from the comparisons in Specific Aim 5 and perform wet-lab experiments to test these hypotheses in ground-based systems and in future spaceflight missions. Expected significance: Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microbes play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses proposed here will attempt to systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight, a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microorganisms play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses performed in this study will systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight--a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress.
Task Progress:	The Specific Aims of this project were: 1. Mine the GeneLab bacterial transcriptome database and reprocess the data in a fashion that will allow cross-organism comparisons. 2. Search the spaceflight literature to identify additional relevant datasets that may have escaped inclusion in the GDS. Deposit these additional datasets in the GDS. Access and reformat the additional data as in Specific Aim 1. 3. Perform bioinformatics analyses to identify pathways and regulons significantly activated or repressed by exposure to spaceflight vs. ground controls in individual organisms. 4. Compare transcriptomes of the same organism grown in simulated microgravity (clinostats), and in actual spaceflight conditions. 5. Perform comparisons across species, genera, families, and higher taxonomic groupings to find similarities and differences in spaceflight responses. 6. Develop specific hypotheses from the comparisons in Specific Aim 5 and perform wet-lab experiments to test these hypotheses in ground-based systems and in future spaceflight missions. In the first year of this grant (FY1), we accomplished Specific Aims 1-3. In the second year (FY2), we accomplished Specific Aim 5. In the third year (No-Cost Extension, FY3) we accomplished Specific Aim 4. Graduate student Michael Morrison

	graduated with his Ph.D. degree in March of 2019. Regarding Specific Aim 6: In Specific Aim 5, we did not find evidence of a common “spaceflight response”; in other words, we did not find any genes expressed in common among all of the species tested. We confirmed this for specific up-regulated spaceflight genes (encoding biofilm and biotin biosynthesis) using qRT-PCR. Because Specific Aim 6 depended on a positive result of Specific Aim 5, there was no longer a reason to pursue Specific Aim 6.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Articles in Other Journals or Periodicals	Morrison MD, Nicholson WL. "Comparisons of transcriptome profiles from Bacillus subtilis cells grown in space vs. High Aspect Ratio Vessel (HARV) clinostats reveal a low degree of concordance." Astrobiology 20:xxx-xxx. (12 p.). In review as of October 2020; expected publication Dec 2020. Future doi will be https://doi.org/10.1089/ast.2020.2235 , Oct-2020
Articles in Peer-reviewed Journals	Morrison MD, Nicholson WL. "Comparisons of transcriptome profiles from Bacillus subtilis cells grown in space vs. High Aspect Ratio Vessel (HARV) clinostats reveal a low degree of concordance." Astrobiology 2020 Dec 14;20(12):1498-509. https://doi.org/10.1089/ast.2020.2235 ; PMID: 33074712 , Dec-2020
Dissertations and Theses	Morrison MD. "Investigating the effects of the human spaceflight environment on Bacillus subtilis." Ph.D. Thesis, University of Florida, May 2019. , May-2019